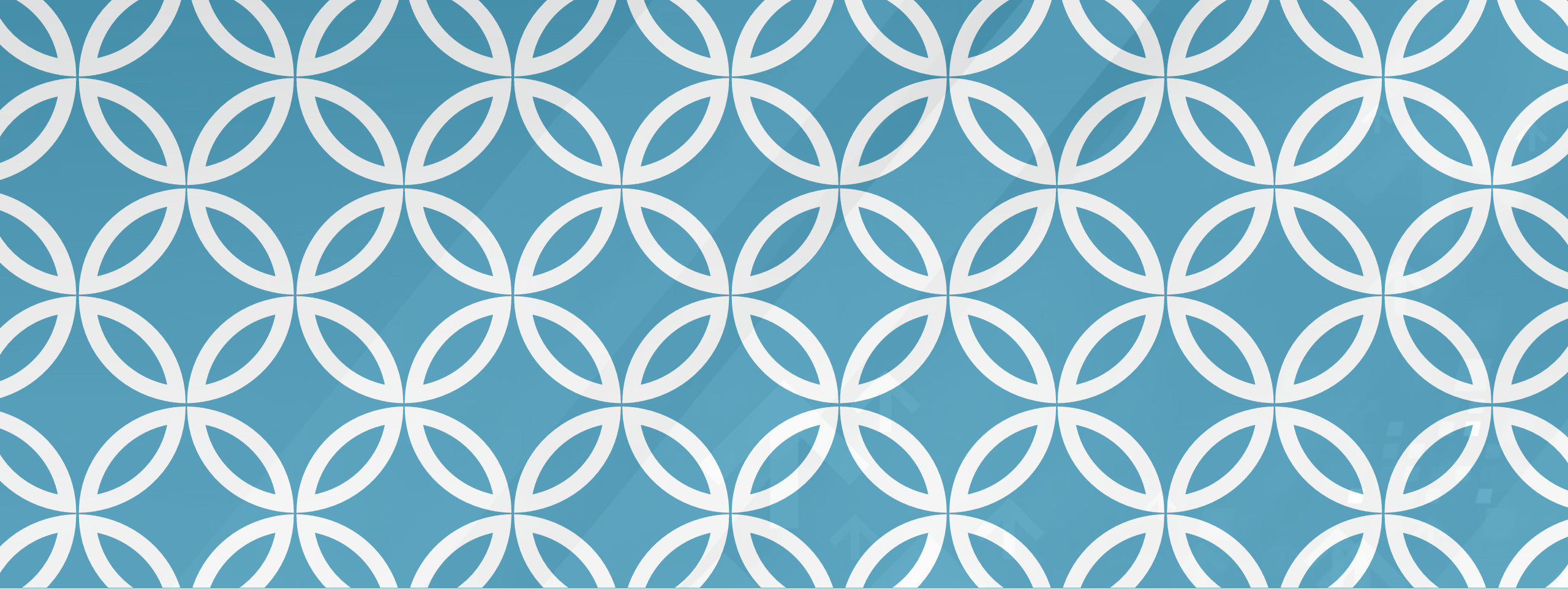




CLUSTERING

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WHAT IS CLUSTERING

Clustering is the classification of objects into different groups, or more precisely, the partitioning of a data set into subsets (clusters), so that the data in each subset (ideally) share some common trait - often according to some defined distance measure.

TYPES OF CLUSTERING

1. **Hierarchical algorithms**: these find successive clusters using previously established clusters.

1. **Agglomerative ("bottom-up")**: Agglomerative algorithms begin with each element as a separate cluster and merge them into successively larger clusters.

2. **Divisive ("top-down")**: Divisive algorithms begin with the whole set and proceed to divide it into successively smaller clusters.

2. **Partitional clustering**: Partitional algorithms determine all clusters at once. They include:

K-means

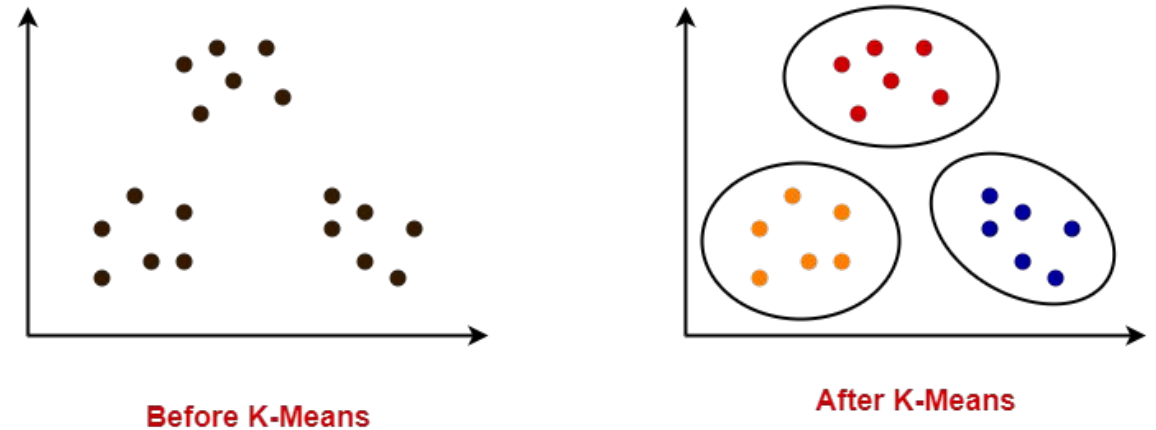
Fuzzy *c*-means clustering

K-MEANS

K-Means clustering is an unsupervised iterative clustering technique.

It partitions the given data set into k predefined distinct clusters.

A cluster is defined as a collection of data points exhibiting certain similarities.



- It partitions the data set such that-
- Each data point belongs to a cluster with the nearest mean.
 - Data points belonging to one cluster have high degree of similarity.
 - Data points belonging to different clusters have high degree of dissimilarity.

K-MEANS CLUSTERING ALGORITHM

1) Choose the number of clusters K.

2) Randomly select any K data points as cluster centers.

Select cluster centers in such a way that they are as far as possible from each other.

3) Calculate the distance between each data point and each cluster center.

The distance may be calculated either by using given distance function or by using euclidean distance formula.

4) Assign each data point to some cluster.

A data point is assigned to that cluster whose center is nearest to that data point.

5) Re-compute the center of newly formed clusters.

The center of a cluster is computed by taking mean of all the data points contained in that cluster.

6) Keep repeating the procedure from Step-03 to Step-05 until any of the following stopping criteria is met-

Center of newly formed clusters do not change

Data points remain present in the same cluster

Maximum number of iterations are reached

SIMILARITY /DISTANCE

1-dimensional similarity/distance



Customer 1	
Age	
54	



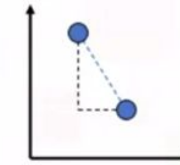
Customer 2	
Age	
50	

$$\text{Dis}(x_1, x_2) = \sqrt{\sum_{i=0}^n (x_{1i} - x_{2i})^2}$$

2-dimensional similarity/distance



Customer 1	
Age	Income
54	190

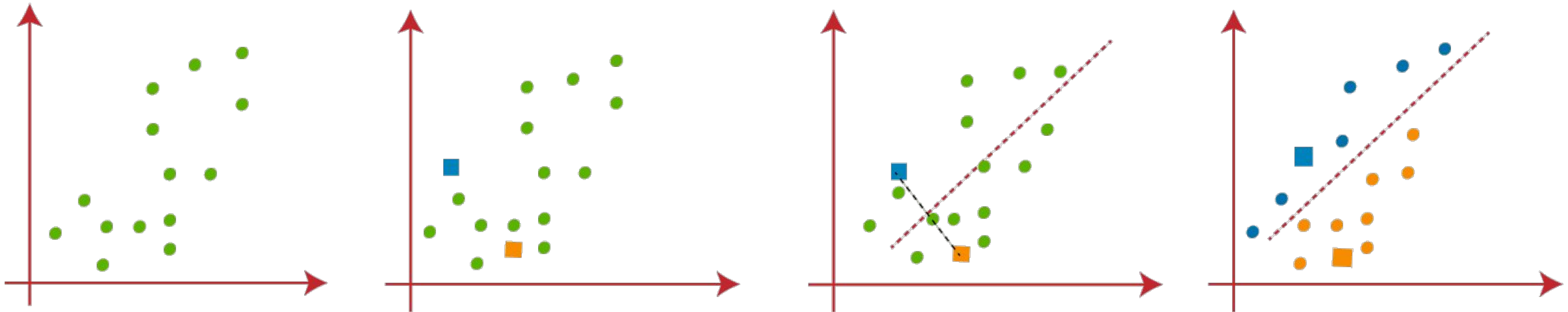


Customer 2	
Age	Income
50	200

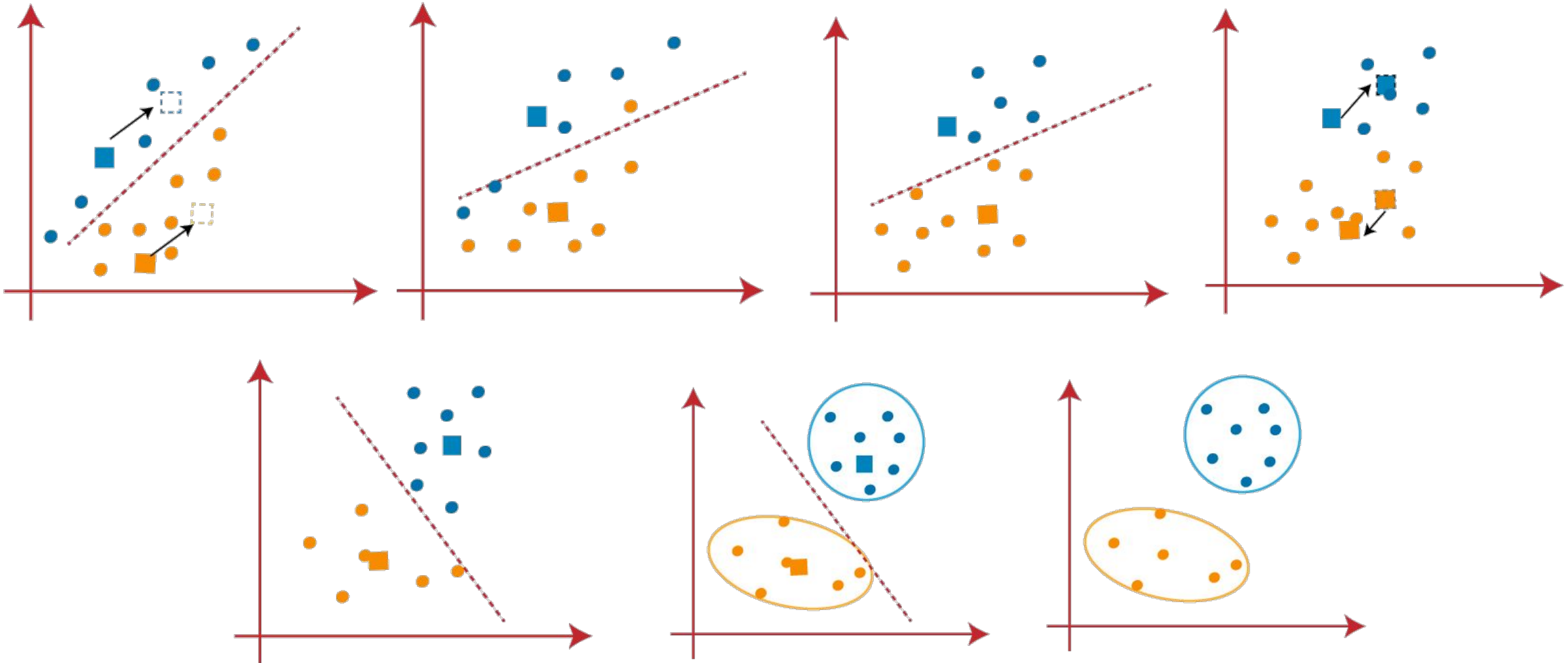
$$\begin{aligned}\text{Dis}(x_1, x_2) &= \sqrt{\sum_{i=0}^n (x_{1i} - x_{2i})^2} \\ &= \sqrt{(54 - 50)^2 + (190 - 200)^2} = 10.77\end{aligned}$$

K-MEANS CLUSTERING ALGORITHM

Suppose we have two variables M1 and M2. The x-y axis scatter plot of these two variables is given below:



K-MEANS CLUSTERING ALGORITHM



HOW TO CHOOSE THE VALUE OF "K NUMBER OF CLUSTERS" IN K-MEANS CLUSTERING?

Elbow Method

The Elbow method is one of the most popular ways to find the optimal number of clusters. This method uses the concept of WCSS value. **WCSS** stands for **Within Cluster Sum of Squares**, which defines the total variations within a cluster. The formula to calculate the value of WCSS (for 3 clusters) is given below:

$$WCSS = \sum_{P_i \text{ in Cluster1}} \text{distance}(P_i C_1)^2 + \sum_{P_i \text{ in Cluster2}} \text{distance}(P_i C_2)^2 + \sum_{P_i \text{ in Cluster3}} \text{distance}(P_i C_3)^2$$

$$\sum_{P_i \text{ in Cluster1}} \text{distance}(P_i C_1)^2$$

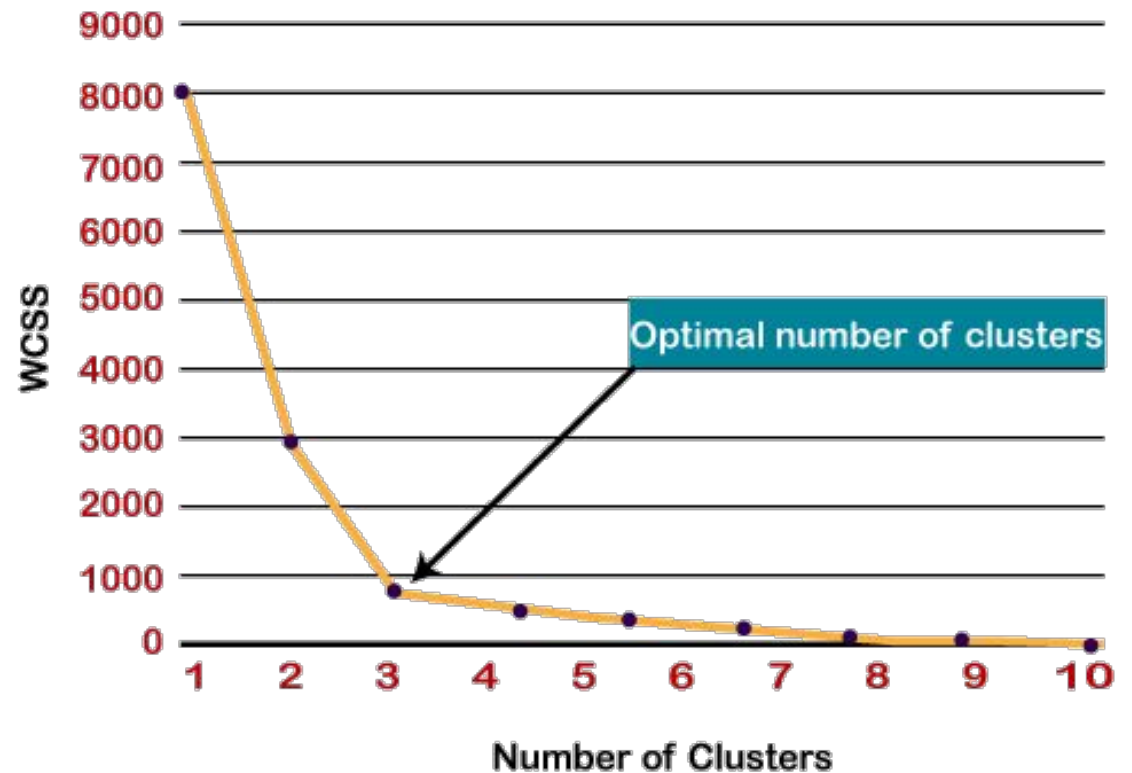
It is the sum of the square of the distances between each data point and its centroid within a cluster1 and the same for the other two terms.

OPTIMAL VALUE OF CLUSTERS

To find the optimal value of clusters, the elbow method follows the below steps:

1. It executes the K-means clustering on a given dataset for different K values (ranges from 1-10).
2. For each value of K, calculates the WCSS value.
3. Plots a curve between calculated WCSS values and the number of clusters K.
4. The sharp point of bend or a point of the plot looks like an arm, then that point is considered as the best value of K.

Since the graph shows the sharp bend, which looks like an elbow, hence it is known as the elbow method.



PROBLEMS BASED ON K-MEANS CLUSTERING ALGORITHM

Cluster the following eight points (with (x, y) representing locations) into three clusters:

A1(2, 10), A2(2, 5), A3(8, 4), A4(5, 8), A5(7, 5), A6(6, 4), A7(1, 2), A8(4, 9)

Initial cluster centers are: A1(2, 10), A4(5, 8) and A7(1, 2).

The distance function between two points $a = (x_1, y_1)$ and $b = (x_2, y_2)$ is defined as-

$$P(a, b) = |x_2 - x_1| + |y_2 - y_1|$$

Use K-Means Algorithm to find the three cluster centers after the second iteration.

SOLUTION

Iteration-01:

We calculate the distance of each point from each of the center of the three clusters.

The distance is calculated by using the given distance function.

The following illustration shows the calculation of distance between point A1(2, 10) and each of the center of the three clusters-

Calculating Distance Between A1(2, 10) and C1(2, 10)-

$$P(A1, C1)$$

$$= |x_2 - x_1| + |y_2 - y_1|$$

$$= |2 - 2| + |10 - 10|$$

$$= 0$$

Calculating Distance Between A1(2, 10) and C2(5, 8)-

$$P(A1, C2)$$

$$= |x_2 - x_1| + |y_2 - y_1|$$

$$= |5 - 2| + |8 - 10|$$

$$= 3 + 2$$

$$= 5$$

Calculating Distance Between A1(2, 10) and C3(1, 2)-

$$P(A1, C3)$$

$$= |x_2 - x_1| + |y_2 - y_1|$$

$$= |1 - 2| + |2 - 10|$$

$$= 1 + 8$$

$$= 9$$

In the similar manner, we calculate the distance of other points from each of the center of the three clusters.

Next,

- We draw a table showing all the results.
- Using the table, we decide which point belongs to which cluster.
- The given point belongs to that cluster whose center is nearest to it.

Given Points	Distance from center (2, 10) of Cluster-01	Distance from center (5, 8) of Cluster-02	Distance from center (1, 2) of Cluster-03	Point belongs to Cluster
A1(2, 10)	0	5	9	C1
A2(2, 5)	5	6	4	C3
A3(8, 4)	12	7	9	C2
A4(5, 8)	5	0	10	C2
A5(7, 5)	10	5	9	C2
A6(6, 4)	10	5	7	C2
A7(1, 2)	9	10	0	C3
A8(4, 9)	3	2	10	C2

From here, New clusters are-

Cluster-01:

First cluster contains points-

- A1(2, 10)

Cluster-02:

Second cluster contains points-

- A3(8, 4)
- A4(5, 8)
- A5(7, 5)
- A6(6, 4)
- A8(4, 9)

Cluster-03:

Third cluster contains points-

- A2(2, 5)
- A7(1, 2)

Cluster-01:

First cluster contains points-

- A1(2, 10)

Cluster-02:

Second cluster contains points-

- A3(8, 4)
- A4(5, 8)
- A5(7, 5)
- A6(6, 4)
- A8(4, 9)

Cluster-03:

Third cluster contains points-

- A2(2, 5)
- A7(1, 2)

Now,

- We re-compute the new cluster clusters.
- The new cluster center is computed by taking mean of all the points contained in that cluster.

For Cluster-01:

- We have only one point A1(2, 10) in Cluster-01.
- So, cluster center remains the same.

For Cluster-02:

Center of Cluster-02

$$= ((8 + 5 + 7 + 6 + 4)/5, (4 + 8 + 5 + 4 + 9)/5)$$
$$= (6, 6)$$

For Cluster-03:

Center of Cluster-03

$$= ((2 + 1)/2, (5 + 2)/2)$$
$$= (1.5, 3.5)$$

This is completion of Iteration-01.

Iteration-02:

- We calculate the distance of each point from each of the center of the three clusters.
- The distance is calculated by using the given distance function.

The following illustration shows the calculation of distance between point A1(2, 10) and each of the center of the three clusters-

Calculating Distance Between A1(2, 10) and C1(2, 10)-

$$\begin{aligned} P(A1, C1) &= |x_2 - x_1| + |y_2 - y_1| \\ &= |2 - 2| + |10 - 10| \\ &= 0 \end{aligned}$$

Calculating Distance Between A1(2, 10) and C2(6, 6)-

$$\begin{aligned} P(A1, C2) &= |x_2 - x_1| + |y_2 - y_1| \\ &= |6 - 2| + |6 - 10| \\ &= 4 + 4 \\ &= 8 \end{aligned}$$

Calculating Distance Between A1(2, 10) and C3(1.5, 3.5)-

$$\begin{aligned} P(A1, C3) &= |x_2 - x_1| + |y_2 - y_1| \\ &= |1.5 - 2| + |3.5 - 10| \\ &= 0.5 + 6.5 \\ &= 7 \end{aligned}$$

In the similar manner, we calculate the distance of other points from each of the center of the three clusters.

Next,

- We draw a table showing all the results.
- Using the table, we decide which point belongs to which cluster.
- The given point belongs to that cluster whose center is nearest to it.

Given Points	Distance from center (2, 10) of Cluster-01	Distance from center (6, 6) of Cluster-02	Distance from center (1.5, 3.5) of Cluster-03	Point belongs to Cluster
A1(2, 10)	0	8	7	C1
A2(2, 5)	5	5	2	C3
A3(8, 4)	12	4	7	C2
A4(5, 8)	5	3	8	C2
A5(7, 5)	10	2	7	C2
A6(6, 4)	10	2	5	C2
A7(1, 2)	9	9	2	C3
A8(4, 9)	3	5	8	C1

From here, New clusters are-

Cluster-01:

A1(2, 10)

A8(4, 9)

Cluster-02:

A3(8, 4)

A4(5, 8)

A5(7, 5)

A6(6, 4)

Cluster-03:

A2(2, 5)

A7(1, 2)

Now,

- The new cluster center is computed by taking mean of all the points contained in that cluster.

Cluster-01:

A1(2, 10)

A8(4, 9)

Cluster-02:

A3(8, 4)

A4(5, 8)

A5(7, 5)

A6(6, 4)

Cluster-03:

A2(2, 5)

A7(1, 2)

For Cluster-01:

Center of Cluster-01

$$= ((2 + 4)/2, (10 + 9)/2)$$

$$= (3, 9.5)$$

For Cluster-02:

Center of Cluster-02

$$= ((8 + 5 + 7 + 6)/4, (4 + 8 + 5 + 4)/4)$$

$$= (6.5, 5.25)$$

For Cluster-03:

Center of Cluster-03

$$= ((2 + 1)/2, (5 + 2)/2)$$

$$= (1.5, 3.5)$$

This is completion of Iteration-02.

After second iteration, the center of the three clusters are-

- C1(3, 9.5)

- C2(6.5, 5.25)

- C3(1.5, 3.5)

DRAWBACKS OF K-MEANS CLUSTERING

The clusters created have a circular shape. This is because the centroids of the clusters are updated iteratively using the mean value.

